A Companion to Aluffi's Algebra Chapter 0

Jiawen Huang

6 January 2026

Acknowledgment

1. Thank Professor Paolo Aluffi for writing such a wonderful book.

2. Main contributors of Discord Discussion Forum:

   Arthur, Evie and Jiawen Huang (AB-BOY)

   (I don't know their real names). Thank Arthur and Evie for discussing many problems with me and helping me understand the book better. I would not be able to finish this companion without their help.

3. Thank Evan Chen for creating such a beautiful latex style.

4. English is not my first language. I apologize for any grammatical errors in this companion (in fact, a lot). Just ignore them. Not that deep.

5. This companion contains my personal notes, results I found beautiful and solutions to most of the challenging exercises. They may not be the best solutions or the most elegant ones. Please feel free to contact me if you have better solutions or suggestions. The discussions in chapter 1 and 2 are sort of meaningless and naive, but I included them for completion.

6. Algebra Chapter 0 is not as difficult as Lang's Algebra. So this companion is not necessary. But I hope it can provide a little help to the readers who are new to abstract algebra or new to college-level math. Readers with more experience may not find this companion useful.

7. I was a high school student when I read the book. I yapped too much, and I am sorry if that bothers you. There may be some mistakes in this companion. Please feel free to contact me if you find any mistakes or have any suggestions.

My website: https://amazingboboy.github.io/my-website/.
My email: kevin47900 [at] gmail [dot] com

Recommendations for readers

- Do not skip any word in the first seven chapters, read it word by word. Aluffi is a gifted writer.

- Please do all the exercises if you are new to abstract algebra. And that was what I did when I was reading the book for the first time. This is important because abstract algebra is very different from other branches of math you may have seen before. It's a language that you need to get used to. Also, there are a lot of interesting results hidden in the exercises, especially in the later chapters! My favorites are the series of exercises about the norm and trace functions in chapter 7. At the end, Aluffi introduces Hilbert's Theorem 90 as an application. This is really cool! (So Hilbert's Theorem 90 basically gives you all the rational solution to Pell's equation if you have learned about it in math competitions before).

- This is a book famous for its early introduction to category theory. But in fact, this book has a lot of other advantages: great exercises, hint, Internal references if you are using pdf version, etc. You will find out as you read more. If one day there is an Algebra chapter 1 by Aluffi, I will be the first to buy it.

Contents

Chapter 1: Preliminaries

This chapter is mainly a review of basic concepts in set theory and category theory. My recommendations for readers:

- Do not skip this chapter, read it word by word. Aluffi is a gifted writer. JUST BE PATIENT and READ. There are many subtle details that you probably have not seen before.
- Try to do all the exercises. Most exercises are straightforward and don't take too much time but really help you solidify the concepts. This is especially important for readers who have not seen category theory before.
- The most important information (in my opinion) you need to take away from this chapter is the universal property. You may think it's useless at first, but I promise it will make sense once you get to section 7 of chapter 2.
- Don't feel bad if you don't feel like you fully understand category theory after reading this chapter. This is normal. It's indeed abstract nonsense if you have no context to apply it to. The problem that makes me realize the power and clarity of category theory is Exercise 7.12 from chapter 2 (prove $F^{ab}(A) \cong F/[F, F]$). You may want to take a look at it when you get there.
- Good Luck!

Exercise 2.5

Discussion. Exercise 2.5

AB-BOY — If f is an epimorphism, then f is surjective.
Not easy.
Its form: for all sets Z and all functions a' and $a'' : B \to Z$, if $a'f = a''f$, then $a' ='' a$.
Proof: $f : A \to B$. Assume f is not surjective for the sake of contradiction, then there msut exist b_0 such that b_0 is not in $Im(f)$. Construct $a'(b_0) = 1$ and $a''(b_0) = 2$, and $a'(Im(f)) = a''(Im(f))$ then $a'f = a''f$ but a' doesn't equal to a''. Contradiction!

Exercise 3.9

Discussion. Exercise 3.9

Evie — I defined MSet like this in 3.9. Did anyone do it differently?

$$\mathrm{Hom}_{\mathbf{MSet}}((A,\sim_A),(B,\sim_B)) := \{f \in B^A \mid \forall x, y \in A.\,(x \sim_A y \implies f(x) \sim_B f(y))\}$$

In this category, a morphism is a monomorphism iff it is a injective, and an epimorphism iff it is surjective, but unlike in Set a morphism that is both mono and epi may not be iso.

Arthur — i did the same thing

AB-BOY — wait, which objects determine ordinary multisets as defined in 2.2 and how what does $(A, \sim)$ mean? i assume A is a regular set and $\sim$ is a equiv relation. Ex. $A = \{a, b, c, d, e\}$ where $a \sim b$, $c \sim d \sim e$ (but a, b, c, d, e are all distinct), then $(A, \sim)$ is essentially a multiset right since it's just (a, a, c, c, c) i understand $A/\sim$ but $(A, \sim)$ is not in the book i think

in my example, are a and b the same thing in $(A, \sim)$?

Arthur — $(A, \sim)$ just means the set A endowed with the equivalence relation $\sim$. In your example, you have a multiset that contains 2 distinct elements: a, which is the same as b, and you have 2 of that element, and c, which is the same as d and e, and you have three of that one.

AB-BOY — $|(A, \sim)|$ is 2 or 5 in this case

Evie — I imagined embedding multisets like a, a, b as sets-with-equivalence-relations $(a, 0), (a, 1), b$ with equivalence relation $(a, 0) \sim (a, 1)$. But there are also objects of MSet like $(a, 0), (a, 1), \ldots$ which would be interpreted as having infinitely many copies of a, which is not a multiset in the "a multiset is a function $A \to \mathbb{N}$" definition
I will say, I do find a lot of the exercises in this book to be worded confusingly, or where it's unclear what the author specifically wants from you. Not the worst-written exercises I've encountered, but could be a lot better

AB-BOY — ok, so the size of $(A, \sim)$ is five

Evie — The size of A is five. The size of $A/\sim$, which corresponds to how many distinct elements the multiset has, is two.

AB-BOY — wait
can you give me an example of a function that is both epi and mono but not iso in MSET
i know this is true in general
like in my opinion, MSet is bascially the same as Set. The objects in MSet are just with an extra relation right?
MSet is isomorphic to SET?

Evie — Let $A = B = 1, 2$ with $\sim_A = (1,1),(2,2)$ and $\sim_B = (1,1),(2,2),(1,2),(2,1)$.
The identity function is a morphism in MSet of $A \to B$ but not $B \to A$, and for $A \to B$ is both epi and mono
This is almost exactly the same situation as there being a continuous function $[0,1) \cup [2,3) \to [0,2)$ that is injective and surjective but has no continuous inverse (and so is both epi and mono in the category of topological spaces with continuous maps, but not iso)

Exercise 4.4

Discussion. Exercise 4.4

AB-BOY — For second question, how can you have a category $C_{nonmono}$ without monomorphism? It must have identity right?

AB-BOY — btw any two non-monomorphism composition will be non-monomorphism.

Arthur — yeah, you can't have such a category, because the identity is always a monomorphism.

Exercise 5.9

Discussion. Exercise 5.9

AB-BOY — Say A, B and C are objects in category $\mathbf{C}$. I think $A \times B \times C$ doesn't make sense in most category. In SET, it makes snese becasue $A \times B$ is also a set. But in general, $A \times B$ may not be the object of $\mathbf{C}$. Then $(A \times B) \times C$ doesn't make sense.

AB-BOY — @Evie
product of objects doesn't make sense sometimes

Evie — "category with products" means a category where any two (or more, including infintely many) objects have a product in the category

5.9. Let C be a category with products. Find a reasonable candidate for the universal property that the product $A \times B \times C$ of *three* objects of C ought to satisfy, and prove that both $(A \times B) \times C$ and $A \times (B \times C)$ satisfy this universal property. Deduce that $(A \times B) \times C$ and $A \times (B \times C)$ are necessarily isomorphic.

Chapter 2: Groups, first encounter

If you have time, please try Exercise 7.12.

Exercise

Discussion. Exercise

AB-BOY —

Arthur —

Evie —

Joke 1.1

Discussion. Joke 1.1

AB-BOY — Groupiod is a category such that all its morphism are isomorphism. How is groupiod a group? I think it would make more sense if we say the $Aut_G(*)$ is a group. And the composition of the groupiod is the operator in the group

Evie — Yeah, the definitions are equivalent.

Exercise 2.12

Discussion. Exercise 2.12

AB-BOY — This is an interesting problem. Technique is infinite descent. Super cool but common in math olympiad.

Exercise 2.15

Discussion. Exercise 2.15

AB-BOY — It's worth thinking about.
Problem itself is super easy. But it brings up a profound question, what is going to happen to the reduce residue system when we change $\mathbb{Z}/n\mathbb{Z}$ to $\mathbb{Z}/np\mathbb{Z}$ where p is a prime.

Evie — This is closely related to 4.9.

Exercise 3.3

Discussion. Exercise 3.3

AB-BOY —

$$\begin{tikzcd}
 & G \arrow[dl, "\pi_g(x)=(x,e_h)"'] \arrow[dr, "f_g"] & \\
G \times H \arrow[rr, "\sigma(x,y)=f_g(x)\cdot f_h(y)" description] & & Z \\
 & H \arrow[ul, "\pi_h(h)=(e_g,h)"] \arrow[ur, "f_h"'] &
\end{tikzcd}$$

I only use the the concept of abelian group once. Is my solution correct? For any $x \in G$ and $y \in H$, $\sigma(x, y) = \sigma((x, e_h) \cdot (e_g, y)) = \sigma((x, e_h)) \cdot \sigma((e_g, y)) = xy = yx$ (here is m the only place i use communicative).

Evie — Looks right to me.

AB-BOY — is it ture when G and H are infinite?
cuz i see some post on mathstack said that this reuslt only holds when G and H are finite. but the book doesn't mention it at all

Evie — Yes it is still true when G and H are infinite "Infinite coproduct" in that link means the coproduct of infinitely many groups.

AB-BOY — oh, i see.
i am scared of "infinite"
but intuitively, this works regardless of the cardinality

Exercise 3.5

Discussion. Exercise 3.5

AB-BOY — I assume author mean $\mathbb{Q}$ is a group with addition. My solution: Assume $\mathbb{Q} = A \times B$ for the sake of contradiction. Assume $1 = (x, y)$ where $x \in A$ and $y \in B$. For any rational $\frac{p}{q}$, it must be some (a, b) where $a \in A$ and $b \in B$. Therefore, $p = q(a, b) = p(x, y)$. Thus, $\frac{a}{b} = \frac{x}{y}$ for all a and b. This is impossible.
Someone check my sol

Evie — Looks right to me. Similar to how I solved it.

II.3.5 $\mathbb{Q}$ is not isomorphic to a direct product

Suppose there were an isomorphism $A \times B \xrightarrow{f} \mathbb{Q}$; then $\langle f((a,0))\rangle$ and $\langle f((0,b))\rangle$ intersect only at $f((0,0)) = 0$. But naming $f((a,0)) = p/q$ and $f((0,b)) = p'/q'$, $(p'q)(p/q) = (pq')(p'/q')$, so the cyclic subgroups have nontrivial overlap in $\mathbb{Q}$.

AB-BOY — make more sense
my solution has a bug, but can be easily fixed.
So neat.

Exercise 4.4

Discussion. Exercise 4.4

Evie — Are $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$ isomorphic?
Has anyone made progress on this yet? I am pretty stuck.
My best idea so far is: Consider both groups as vector spaces over $\mathbb{Q}$. Both have dimension $|\mathbb{R}|$, so there is a bijection between their bases that should extend to a bijective linear map from $\mathbb{R}$ to $\mathbb{C}$, and a linear map is just a group homomorphism with extra structure (respects scalar multiplication). But this does not seem like the kind of proof that's intended, and I also am not really familiar with the details for infinite-dimensional vector spaces (does same dimension imply linear isomorphism, or is that only for finite-dimensional vector spaces?)

Arthur — i think that assuming the axiom of choice, your proof works (just pick a basis for both, map one to the other, extend linearly). but yeah, i don't think this is the intended solution at this point in the book, especially given that problem IV.1.1 is proving that RR and CC are QQ-linearly isomorphic

AB-BOY — it makes sense to me if both of them have the same dimension oops, I am also unfamiliar with infinite dimension. I hate infinite. These things are counterintuitive.

Exercise 4.12

Discussion. Exercise 4.12

AB-BOY — $x^3 = 9$ ahs no solution because $1 = (x^3)^{10} = 9^{10} \neq 1 \pmod{31}$

Exercise 4.13

Discussion. Exercise 4.13

AB-BOY — first of all, $(0,0)$ must go to $(0,0)$. For the rest of the three distinct elements a, b and c in $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. They satisfy a key fact: $a + b = c$. This makes all permutations a group homomorphism. Just to share some thoughts
$a + a$ is always $(0,0)$
a little cute structure

Exercise 4.15

Discussion. Exercise 4.15

AB-BOY — compute the automorphism of $(\mathbb{Z}, +)$.

So it must satisfy $f(a) + f(b) = f(a+b)$. THIS IS Cauchy equation in functional equation. So the solution in $\mathbb{Z}$ is $f(x) = kx$ But since we want an isomorphism so k is either 1 or -1.

Evie — Another solution is just to observe that a homomorphism with $\mathbb{Z}$ (or any of its quotients) is fully determined by where it sends 1.

Exercise 5.4

Discussion. $w \cdot w' = R(w \cdot w')$ is associative.

Evie — I was stuck on this one. Of course there is the brute-force approach but surely there is something better

AB-BOY — ?brute force?
wut
how
there are infinitely many cases i mean
did u know any nice approach?
i mean induction is fine
wait
actually
induction is very straightforward
it's kinda hard to type the proof let me know if you want to know my proof, we can call (i want to share lol, if you are available)
@Arthur i have a solution that is pretty neat i think, using induction
The idea is to induce on the length of the words
Hi Evie

Evie — I think there is a more beautiful proof from the fact that multiplication in the free group is just concatenation of paths in the Cayley graph (with cycles removed), and concatenation of paths is obviously associative
I'm not sure of the details yet though

AB-BOY — Cayley graph in n dimension? Sounds interesting

Arthur — i dont see how you induce on the length of the words

AB-BOY — Assume the claim is true for n
In n+1, consider the last element a.
Case 1: the nth element is not a^{-1}
Case 2: the nth element is a^{-1}

Arthur — okay, that's essentially where I was stuck. Case 2 does not seem obvious to me

So I basically explain it over a call, and he agreed with my solution.

Exercise 5.6

Discussion. Exercise 5.6

AB-BOY — when author asks me to prove G is a coproduct of A and B in Grp, does he want me to draw the catgeory diagram and find a unqiue homomorphism (prove the G is an initial in the $Grp^{A,B}$)

Arthur — Yeah

Exercise 6.7

Discussion. Exercise 6.7

AB-BOY — if Aut(G) is cyclic, then G is abelian. I found it difficult to prove this amazing result if you do not know Inn. I appreciate how author guide us from Inn to Aut. This is a fantastic proof.
when we study a large group, we should look into some of its subgroups. it's like when we solve a Diophantine equation, it's hard to solve it directly. mod 3,4,5 etc. make our life easier.

Exercise 6.9

Discussion. Exercise 6.9

AB-BOY — someone checks my solution please.

So for any finitely generated subgroup G of $\mathbb{Q}$, there must exists a surjective homomorphism ϕ from $\mathbb{Z}^{\oplus n}$ to G. $n = 1$ is trivial. Let's look at the case when $n = 2$. So for any $x \in G$, there must exists integers a and b such that $x = a \times \phi(1, 0) + \times\phi(0, 1)$. Assume $\phi(1, 0) = \frac{p}{q}$ and $\phi(0, 1) = \frac{m}{n}$. Then $x = \frac{apn+bmq}{qn}$. Apply Bezout's theorem, there must exists a_0 and b_0 such that $a_0pn + b_0mq = gcd(pn, mq) = d$. So $\frac{d}{qn}$ will be the generator of G. Thus, G is cyclic.

This solution starts from the categorical definition. but in this problem, using the latter definition is easier.

AB-BOY — nvm
for any finitely generated group, let's assume its generators are a_1 to a_n. there must exists a k such that $k \mid a_i$ is an integer. So $< a_1, \cdots, a_n >$ is a subgroup of $< k >$. Thus, $< a_1, \cdots, a_n >$must be a cyclic group

Evie — Reading your solution now. There is also the option to just multiply everything in the subgroup by the lcm of the denominators of the generators to get a subgroup of $(\mathbb{Z}, +)$, show that subgroup is cyclic, and divide by the lcm again
Oh didn't see your second post, which is basically that.
Yes, it looks right to me in the 2-generator case
Doesn't affect the validity, but "Bezout's theorem" is a theorem in algebraic geometry that generalizes "polynomials of degree N intersect at at most N points". The one you mean is called Bezout's lemma.

Exercise 6.11

Discussion. Exercise 6.11

AB-BOY — wut? coproduct of a non abelian group must be infinite right? so not S_3

Evie — Yeah that should work

Exercise 7.7

Discussion. Exercise 7.7

AB-BOY — Not an easy one but solved! Key 1: if x is an element of order n, then for any $g \in G$, $(gxg^{-1})^n = e$

Key 2: if $x = h_1...h_k$ is in H where h_i has order of n, $gxg^{-1} = (gh_1g^{-1})(gh_2g^{-1})...gh_kg^{-1}$

Exercise 7.11

Discussion. Exercise 7.11

AB-BOY — Key: $gaba^{-1}b^{-1}g^{-1} = (gag^{-1})(gbg^{-1})(ga^{-1}g^{-1})(gb^{-1}g^{-1}) = (gag^{-1})(gbg^{-1})(gag^{-1})^{-1}(gbg^{-1})^{-1}$
Motivation: Define $\gamma_g = f$. Note that $f(ab) = f(a)f(b)$ THIS IS A CRAZY FUNCTION! And f is a homomorphism. Favorite function in this book
Thus, $f(aba^{-1}b^{-1}) = f(a)f(b)f(a^{-1})f(b^{-1}) = f(a)f(b)f(a)^{-1}f(b)^{-1}$

Evie — Thanks. I feel like I tried a bunch of things like this, so I'm not sure how I missed it

AB-BOY — yeayea $G/[G,G]$ is abelian $\Leftrightarrow ab \sim ba \Leftrightarrow ab(ba)^{-1} = aba^{-1}b^{-1} \in [G,G]$ which is obvious

Evie — Anyway I always think of γ_g as "do something in a different reference frame", like when you diagonalize a matrix.

Exercise 8.1

Discussion. Exercise 8.1

AB-BOY — This is true right?
$G_1 \cong \frac{G_1 \times H}{e \times H} \cong \frac{G_1 \times H}{H \times e} \cong \frac{G_1}{H} \times \frac{H}{e} \cong \frac{G_2}{H} \times \frac{H}{e}$

—

Arthur — No. $S_3/C_3 = C_2 = C_6/C_3$ but C_6 and S_3 are different, as per examples 8.5 and 8.6

AB-BOY — what was wrong about my sol?

Arthur — im not sure you can split the quotients as you do $\frac{G\times H}{H\times e} = (G/H) \times (H/e)$
everything else checks out i think

AB-BOY — Claim 8.4.

Arthur — ah yea
H is not necessarily normal though

AB-BOY — wut? then G/H is not a group

Arthur — ah yeah.

AB-BOY — in this example, C_3 is normal

Arthur — yeah as well

AB-BOY — wait if $X \cong Y$ and X and Y are both normal in G. does it imply $G/X \cong G/Y$

Arthur — no i dont think so

AB-BOY — that's not necessarily true? X and Y can be dif subgroup of G
i think that's the problem

Arthur — $\mathbb{Z}/d\mathbb{Z}$ are all different but all $d\mathbb{Z}$ are isomorphic to $\mathbb{Z}$

AB-BOY — $\frac{G_1\times H}{e\times H} \cong \frac{G_1\times H}{H\times e}$ is invalid.
$\mathbb{Z}$ and $d\mathbb{Z}$ are perfect counterexample here.

AB-BOY — yeah ok but the reason your proof didn't work was very subtle

Exercise 8.3

Discussion. Exercise 8.3

AB-BOY — this is a cool result
how do you guys prove it?
my proof: choose the set A to be the underlying set of group G. Assume the underlying set of G contains n element: a_1 to a_n. for any i and j, $a_i \cdot a_j = f(i, j)$

Choose R to be the **normal subgroup** generated by

$$\{a_i a_j f(i, j)^{-1} \mid \text{for all } i \text{ and } j\}$$

we know they there must exits a homomorphism from F(G) to G (cuz F(G) is an initial) and the kernel is K.
we need to prove $K = R$
so for any element in R, it's clearly in K
so for any element in R, it's clearly in K
for any word in K, assume it's length is n. we induce on n. $n = 1, 2$ are trivial cases.
when $n = m + 1$, assume the first two element is a_1 and a_2. Assume the word is $a_1 a_2 Z$ where Z contains $m-1$ letters. We know that $a_1 a_2 f(1, 2)^{-1}$ can be used. By assumption, we construct the word $f(1, 2)^{-1} Z$ since it's in the kernel and with length $\leq n$
So we conclude that $K = R$
so any finite group is finitely presneted. guys check my proof
also i want to know how you guys apporach this

Arthur — not sure i understand the details of your proof but i used the same general idea.

Problem 8.3: Prove that every finite group is finitely presented.

Solution: Let G be a finite group, and let e be the unit of G. Let $F = F(G \setminus \{e\})$ be the free group over the finite set $G \setminus \{e\}$. Let R be the set of relations on F defined by

$$R = \{xy \mid x, y \in G \setminus \{e\}, x = y^{-1}\} \cup \{xyz^{-1} \mid x, y, z \in G \setminus \{e\}, xy = z\}.$$

Let N be the normal closure of R in F. We want to show that $G \cong F/N$.

The universal property of the free group for F gives us a group homomorphism $\varphi : F \to G$ making the following diagram commute:

$$\begin{tikzcd} F \arrow[r, dashed, "\varphi"] & G \\ G \setminus \{e\} \arrow[u, "i"] \arrow[ur, "i"'] & \end{tikzcd}$$

where i is the injection. The group homomorphism φ is clearly surjective, we only need to show that $\ker \varphi = N$.

We'll use the following claim:

Claim: A word $w = a_1 \cdots a_n$ in F is in $\ker \varphi$ if and only if the product $a_1 \cdots a_n = e$ in G.

- $\ker \varphi \subseteq N$: We prove by induction on the length of a word $w \in F$ that $w \in \ker \varphi \Rightarrow w \in N$. The only word of length 0 is the empty word, which is both in $\ker \varphi$ and N, so OK. By definition, there is no word of length 1 in $\ker \varphi$, so OK as well. If $|w| \geq 2$, write
$$w = \underbrace{a_1 a_2}_{:=x} \cdots a_n.$$
There are now two cases:
  - $x = e$. In this case, $a_1 a_2 \in R \subset N$, and $a_3 \cdots a_n = e$, so by induction $a_3 \cdots a_n \in N$. Therefore, as a product of two elements of N, $a_1 \cdots a_n \in N$.
  - $x \neq e$. In that case there is a letter y such that, $a_1 a_2 y \in R$ and $xy \in R$ and by induction, $x a_3 \cdots a_n \in N$. Then
$$a_1 \cdots a_n = \underbrace{a_1 a_2 y}_{\in N} \underbrace{(xy)^{-1}}_{\in N} \underbrace{x a_3 \cdots a_n}_{\in N} \in N.$$

  We conclude by induction.
- $N \subseteq \ker \varphi$: Since $\ker \varphi$ is normal, it is enough (and trivial) to show that $R \subset \ker \varphi$. Since N is the normal closure of R, this works.

AB-BOY — basically the same yeah
but u don't have to exclude the identity right?

Arthur — yeah i dont think so. it just seemed cleaner to me

Exercise 8.6

Discussion. Exercise 8.6

AB-BOY — such a nice result but easy to prove. if the graph is a tree, then generators of group has no relationship, then it must be free

Exercise 8.8

Discussion. Exercise 8.8

AB-BOY — Answer: (R^*, x)?

Evie — Yeah, this was my answer
Proof: The determinant is a surjective group homomorphism $GL_n \to \mathbb{R}^*$, and its kernel is SL_n. First isomorphism theorem says that $\mathbb{R}^*$ is isomorphic to GL_n/SL_n

Exercise 8.9

Discussion. Exercise 8.9

Evie — How did you solve this? I really feel like there should be a beautiful solution but I can't find it.

AB-BOY — U send the matrix in Ex6.3 to Exercise 8.9. Same a, b, c, d That is actually a homomorphin I checked it by computer Surprising I searched it up online, it has something to do with quaternion which I am not familiar with I also feel like I need to learn more linear alg ngl I never really handle complex matrix Like I know most of the defs but need to learn more about it

The multiplication of the given matrix with parameters a, b, c, d and the similar matrix with parameters p, q, r, s results in a rotation matrix corresponding to the quaternion product. The product matrix is:

$$\begin{pmatrix} w^2+x^2-y^2-z^2 & 2(xy-wz) & 2(xz+wy) \\ 2(wz+xy) & w^2-x^2+y^2-z^2 & 2(yz-wx) \\ 2(xz-wy) & 2(wx+yz) & w^2-x^2-y^2+z^2 \end{pmatrix}$$

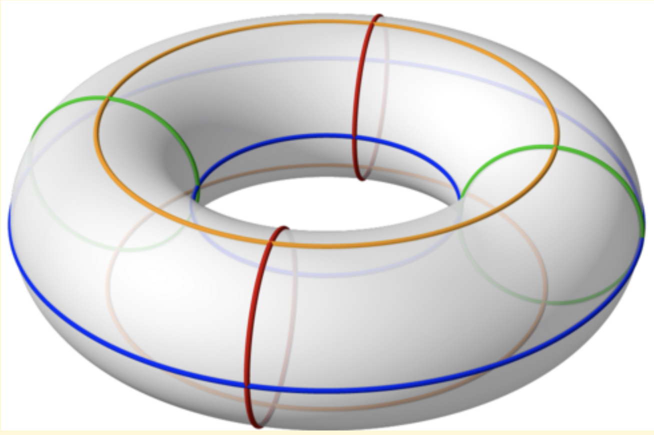
where:

- $w = ap - bq - cr - ds$
- $x = aq + bp + cs - dr$
- $y = ar - bs + cp + dq$
- $z = as + br - cq + dp$

Exercise 8.10

Discussion. Exercise 8.10

AB-BOY — What is $\frac{\mathbb{R}\times\mathbb{R}}{\mathbb{Z}\times\mathbb{Z}}$ is it a Torus?



Evie — Yes it is.

Exercise 8.14

Discussion. Exercise 8.14

AB-BOY — quite interesting! $f : g \to g^k$. Construct a integer E such that $E \equiv 0 \pmod{k}$ and $E \equiv 1 \pmod{n}$ By Chinese remineder theorem, E must exist.
For any $x \in G$, we take $f(x^{\frac{E}{k}}) = x^E$.
Since the order of x, say r, must divide n. So $E \equiv 1 \pmod{r}$. So $x^E = x^{mr+1} = x$.
Thus, f is surjective.
This is super useful.
Ex. When p is a prime such that $p \equiv 2 \pmod{3}$. $x^3 = a \pmod{p}$ always have a solution. Because $p-1$ is coprime to 3!

Evie — Minor stylistic complaint: $\to$ is be used to describe the signature of a function. $\mapsto$ is the one you meant, ie. $f : G \to G : g \mapsto g^k$.

AB-BOY — Ok bud.

Arthur — interesting, i didn't use the chinese remainder theorem. Instead, since G is finite, I show that the function is injective: take a, b two integers such that $ak + bn = 1$. If g and h are two elements of the group such that $g^k = h^k$, since $g^n = e = h^n$, we have that $g = g^{ak+bn} = h^{ak+bn} = h$, so the function $x \mapsto x^k$ is injective and therefore surjective.

AB-BOY — amazing solution! Bezout lemma.

Exercise 8.17

Discussion. Exercise 8.17

AB-BOY — Normality is such a powerful condition. Think about it.

For a normal subgroup H.

$aH \times bH$. Intuitively, it should have about $|H|^2$ (i mean, we really don't know) element right? But since H is a subrgoup and it's normal. It tells us $aH \times bH = abb^{-1}HbH = ab(b^{-1}Hb)H = abHH = abH$. THIS IS A SUPER POWER TOOL. $(xH)^p$ is the same as x^pH. I know it's very obvious when you consider it in the view of equivlent relationship. But When I look at $aH \times bH = abH$ without considering relationship and quotient group, I finally realize how powerful it is.

In this problem, the indcutive step reuqires this insight. If there is a element x of order p in G/H, then $(xH)^p = H$ in G. So there musts exist an $h_0 \in H$ such that $(xh_0)^p = e$.

Exercise 9.4

Discussion. Exercise 9.4

AB-BOY — what is the another natural right action author wants my idea: ag^{-1}

Arthur — i used the conjugation in reverse $g^{-1}hg$

AB-BOY — it's not a right action? wait idk man, that may work it's an action anyway...

Arthur — pretty sure it is. but in any case, i'm not sure there's a right answer here, there's plenty of examples

Evie — My answer was also $g^{-1}hg$. Another is $\alpha(g, h) = g^{-1}h$.

AB-BOY — i am confused about left and right group action. so in my opinion, there are no left and right. it's just a homomorphism σ from G to Aut(A). We abbreviate $\sigma(g)(a) = ga$ I mean it has nothing to do with left and right? What does left and right action actually mean, can someone give me an example of right action?
σ is just a function. it just convention that human being write function on the left and variable on the left?
weird. i just ignore the words "right" and "left" when i read the text.

Arthur — a right action is not a group homomorphism from G to Aut(A) if you write your right action as $\rho(g, a) = ag$, then you have $\rho(gh, a) = agh = \rho(h, \rho(a, g))$

AB-BOY — ? so what is a right action i mean i know what's an action of group on a set of a category

Arthur — i think it's a function ϕ from G to Aut(A) such that $\phi(gh) = \phi(h)\phi(g)$

AB-BOY — wait so u are saying right action is not a homomorpshim ??? so right action is not an action left action= action but right action is somehting else.
so from ur point of view, right action is a function ϕ from G to Aut(A) such that $\phi(gh) = \phi(h)\phi(g)$

Arthur — yeah

This discussion made me understand what right group action is.

Exercise 9.9

Discussion. Exercise 9.9

AB-BOY — i am not sure about my solution:
Key 1: Any set can be partitioned into serveral orbits and actions on orbits are all transitive
Key 2: each orbit will be isomorphic to a $G/Stab(a)$, where a is an element in orbit
Key 3: orginal set A= the coproduct of all the orbits = coproduct of all $G/stab(a)$

Arthur — yup i have the same thing

Exercise 9.10

Discussion. Exercise 9.10

AB-BOY — wut?? $|G/H| = \frac{|G|}{|H|} = |H/G|$ so there is a bijection???
also, i don't know how to use proposition 9.9. ho do u find a set A such that A is acted transitively by G and the $stab(a)$ is equal to H. Or given a subgroup H of G, does there always exist a set A (be acted transitively by G) such that $stab(a) = H$

Evie — I also didn't use the hints. My approach was that a right-coset of G is just a left-coset of the opposite group, and G is isomorphic to its opposite group

Arthur — the proof should work even if the group is infinite.

Evie — I think Lagrange's theorem still works if you define division of infinite cardinals correctly, even if it's much less useful

Arthur — ah yeah you're right i hadn't thought about that

Evie — Lagrange's theroem is that $|G| = [G : H] \times |G/H|$. When —G— is infinite this is the same as $|G| = \max([G : H], |G/H|)$, so for infinite groups we would have $\max([G : H], |G/H|) = \max([G : H], |G\backslash H|)$. I don't think this is enough to guarantee that $|G/H| = |G\backslash H|$ without additional assumptions (like $[G : H]$ being finite)

Arthur — yeah. you can still probably reason using only cardinals, but it's kinda ugly.

Exercise 9.13

Discussion. Exercise 9.13

AB-BOY — interesting.
First, I don't realize gHg^{-1} i a subgroup of G although it's trivial. Second, $\phi : H \to (gHg^{-1}$ is not easy to find.
my phi :$\phi(xH) = (xg^{-1})(gHg^{-1})$ it's indeed bijective

Exercise 9.14

Discussion. Exercise 9.14

Evie — I've been working on this for a few days and I'd love to see how you guys are solving it. I have some ideas but they are all either very ugly or fruitless.

AB-BOY — Yes yeah! My favorite problem of the section
for $(y^{\pm}x)(y^{\pm}x)\cdots(y^{\pm}x)y^{\pm}$. Take $r<0$, th result must be positive, contradiction.
For $x(y^{\pm}x)(y^{\pm}x)\cdots(y^{\pm}x)$, take r to be positive, the result must be neagtive
becasue $x(r)=\frac{-1}{r}$
for $(y^{\pm}x)(y^{\pm}x)\cdots(y^{\pm}x)$ we do case work.
Case 1: $(y^{\pm}x)(y^{\pm}x)\cdots(yx)$ (the last pair is (yx)). We take $r=-\frac{\sqrt{2}}{2}$ (any -1¡irrational¡0). yx(r) must be in positive. So it will be positive forever, which contradict with the fact that $r<0$
Case 2: $(y^{\pm}x)(y^{\pm}x)\cdots(y^{-1}x)$ (the last pair is $y^{-1}x$). we take $r=-\sqrt{2}$ (or any irrational ¡-1). $y^{-1}x(r)$ must be positive. So it will be positive forever
agian, contradiction
another fantastic fact.

$$M=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

such that $a+d=0$, then $M^2=I$ (identity)
when $M\in PSL_2((Z))$
When $det(M)=(a+d)^2$, $M^3=I$ (i am not sure about this one)

Evie — Hm, so the idea is that all sufficiently negative inputs to $y^{-1}x$ become positive, and multiplying by/composing with yx doesn't solve this problem, so no product of $y^{-1}x$ and yx can give the identity. I like that, it's very pretty
and $x(y^{\pm1}x)\cdots(y^{\pm1}x)$ still has that sign problem
And then y and y^{-1} both swap the sign of r for sufficiently large $|r|$, so if you put one of them on the right side then you'll have the problem that sufficiently positive values become negative

AB-BOY — Wait
My solution includes this case
Do u mean it also has a sign problem which leads to contradiction?

AB-BOY — This result is somehow related to an AMC problem: 2024 AMC 12A Problem 25

Exercise 9.16

Discussion. Exercise 9.16

AB-BOY —

$$\begin{tikzcd}
H \arrow[r, hook, "i"] \arrow[dr, "\sigma \circ i"'] & G \arrow[d, "\sigma"] \\
& \text{Aut(vertices of the Tree)}
\end{tikzcd}$$

So H must act on the tree. Since G act on the tree freely, H must act on the tree freely. Thus, H must be a free group.
someone checks my solution And i am not too sure how to prove: only free group act freely on tree.

Arthur — yeah i have the same solution. and i expect that the proof of "only free groups act freely on trees" is pretty hard and uses notions we don't know yet.

Exercise 9.17

Discussion. Exercise 9.17

AB-BOY — a functional equation.
For any $\phi \in \text{Aut}_{G-set}(G)$, $\phi(xy) = y\phi(x)$ for any x and y in G. Take $x = e$. $\phi(y) = y\phi(e)$.
So the value of $\phi(e)$ uniquely determines the ϕ. So $\text{Aut}_{G-set}(G) \cong G$

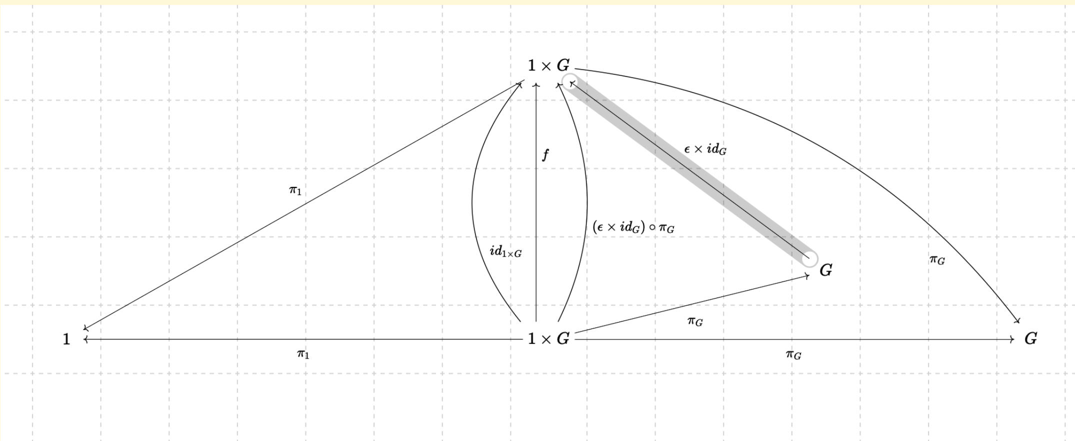
Exercise 10.1

Discussion. Exercise 10.1

AB-BOY — $1 \times G \to G \to 1 \times G$ is an identity
first of all, since $1 \times G$ is a product in $C_{1,G}$. so there must exists π_1 and π_G such that $1 \times G$ with π_1 and π_G is the final in $C_{1,G}$.
Based on the diagram, there must exists morphism $\epsilon \times id_G$ from G to $1 \times G$

$$\begin{tikzcd}
 & 1 \times G \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_G"] & \\
1 & & G \\
 & G \arrow[ul, "\epsilon"] \arrow[ur, "id_G"'] \arrow[uu, "\exists! \epsilon \times id_G" description] &
\end{tikzcd}$$

From the previus diagram, we know that $(\epsilon \times id_G) \circ \pi_G = id_G$. Now, we show that $\pi_G \circ (\epsilon \times idG) = id1 \times G$
By the universal property of product, there exists a unique, say f morphism from $1 \times G$ to $1 \times G$ with the morphism π_1 and π_G such that the diagram commutes.
First, we prove the $f = id_{1 \times G}$ this is trivial
Now, we prove that $f = id_{1 \times G} = \pi_G \circ (\epsilon \times id_G)$. It suffices to show that $\pi_G \circ (\epsilon \times id_G)$ makes the diagram commutes
It suffices to show that (1) $\pi_1 \circ [\pi_G \circ (\epsilon \times id_G)] = \pi_1$ and (2) $\pi_G \circ [\pi_G \circ (\epsilon \times id_G)] = \pi_G$



(2) is trivial !
FOr (1), note that 1 is the final in category. There exists a unqiue morphism from $1 \times G$ to 1. So $\pi_1 \circ [\pi_G \circ (\epsilon \times id_G)]$ must be π_1, by the universal property of 1. We're done!

Exercise 10.5

Discussion. Exercise 10.5

AB-BOY — wow. A group object in Grp is an abelian group! inverse function is homomorphism

Exercise

Discussion. Exercise

AB-BOY —

Arthur —

Evie —